Definición: Convolución

Si las funciones $f \quad y \quad g$ son continuas por tramos en $[0,+\infty[$, entonces:

$f * g = \int_0^t f(\tau) g(t-\tau) d\tau$ se llama la **convolución** de f y g.

Ejemplo :

a) Hallar $e^t * sen(t)$

Solución

Haciendo $f(t) = e^t \quad y \quad g(t) = sen(t)$

$$f * g = \int_0^t e^{\tau} sen(t-\tau) d\tau$$

$$f * g = \int_t^0 e^{t-u} sen(u)(-du) \quad \text{, con } u = t-\tau$$

$$f * g = \int_0^t e^{t-u} sen(u) du$$

$$f * g = \int_0^t e^t e^{-u} sen(u) du$$

$$f * g = e^t \int_0^t e^{-u} sen(u) du$$

$$f * g = e^t \left[\frac{e^{-u}}{2}(-sen(u) - \cos(u)) \right]_0^t$$

$$f * g = e^t \left[\frac{e^{-t}}{2}(-sen(t) - \cos(t)) - \frac{e^0}{2}(-sen(0) - \cos(0)) \right]$$

$$f * g = e^t \left[\frac{e^{-t}}{2}(-sen(t) - \cos(t)) - \frac{1}{2}(-1) \right]$$

$$f * g = \frac{1}{2}(-sen(t) - \cos(t)) + \frac{1}{2} e^t$$

$$f * g = -\frac{1}{2} sen(t) - \frac{1}{2}\cos(t) + \frac{1}{2} e^t$$

b) Encontrar $\mathscr{L}\{e^t * sen(t)\}$

Solución

En el ejercicio anterior encontramos $e^t * sen(t)$, luego

$$\mathscr{L}\{e^t * sen(t)\} = \mathscr{L}\left\{-\frac{1}{2} sen(t) - \frac{1}{2}\cos(t) + \frac{1}{2} e^t\right\}$$

$$= -\frac{1}{2}\mathscr{L}\{sen(t)\} - \frac{1}{2}\mathscr{L}\{\cos(t)\} + \frac{1}{2}\mathscr{L}\{e^t\}$$

$$= -\frac{1}{2}\left(\frac{1}{s^2+1}\right) - \frac{1}{2}\left(\frac{s}{s^2+1}\right) + \frac{1}{2}\left(\frac{1}{s-1}\right)$$

$$= \frac{-(s-1) - s(s-1) + (s^2+1)}{2(s^2+1)(s-1)}$$

$$= \frac{-s+1-s^2+s+s^2+1}{2(s^2+1)(s-1)}$$

$$= \frac{2}{2(s^2+1)(s-1)}$$

$$= \frac{1}{(s^2+1)(s-1)}$$

NOTA: La convolución tiene la propiedad de conmutatividad $\boxed{f*g=g*f}$

Teorema: **(De La Convolución)**

Si $f(t)$ y $g(t)$ son continuas por tramos en $[0,+\infty[$ y de orden exponencial, entonces

$$\begin{aligned}\mathscr{L}\{f*g\} &= \mathscr{L}\{f(t)\}\mathscr{L}\{g(t)\} \\ &= F(s)G(s)\end{aligned}$$

Ejemplo: Hallar $\mathscr{L}\{e^t * sen(t)\}$

Solución

$$\mathscr{L}\{e^t * sen(t)\} = \mathscr{L}\{e^t\}\mathscr{L}\{sen(t)\} = \left(\frac{1}{s-1}\right)\left(\frac{1}{s^2+1}\right)$$

$$\mathscr{L}\{e^t * sen(t)\} = \frac{1}{(s^2+1)(s-1)}$$

Forma Inversa Del Teorema De La Convolución

$$\begin{aligned}\mathscr{L}^{-1}\{F(s)G(s)\} &= \mathscr{L}^{-1}\{F(s)\} * \mathscr{L}^{-1}\{G(s)\} \\ &= f(t)*g(t)\end{aligned}$$

Ejemplo: Hallar $\mathscr{L}^{-1}\left\{\frac{s-1}{(s+1)\left[(s-1)^2+1\right]}\right\}$

Solución

$$\mathscr{L}^{-1}\left\{\frac{s-1}{(s+1)\left[(s-1)^2+1\right]}\right\} = \mathscr{L}^{-1}\left\{\left(\frac{s-1}{(s-1)^2+1}\right)\left(\frac{1}{s+1}\right)\right\}$$

$$= \mathscr{L}^{-1}\left\{\frac{s-1}{(s-1)^2+1}\right\} * \mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}$$

$$= \mathscr{L}^{-1}\left\{\left.\frac{s}{s^2+1}\right|_{s\to s-1}\right\} * e^{-t}$$

$$= e^t\,\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\right\} * e^{-t}$$

$$\mathscr{L}^{-1}\left\{\frac{s-1}{(s+1)\left[(s-1)^2+1\right]}\right\} = e^t\cos(t) * e^{-t}$$

Si hacemos $f(t)=e^{-t}$ y $g(t)=e^t\cos(t)$

$$e^{t}\cos(t)*e^{-t}=\int_0^t e^{\tau}\cos(\tau)e^{-(t-\tau)}d\tau$$

$$e^{t}\cos(t)*e^{-t}=\int_0^t e^{\tau}\cos(\tau)e^{-t+\tau}d\tau$$

$$e^{t}\cos(t)*e^{-t}=\int_0^t e^{2\tau}\cos(\tau)e^{-t}d\tau$$

$$e^{t}\cos(t)*e^{-t}=e^{-t}\int_0^t e^{2\tau}\cos(\tau)d\tau$$

$$e^{t}\cos(t)*e^{-t}=e^{-t}\left[\frac{e^{2\tau}\left(2\cos(\tau)+sen(\tau)\right)}{2^2+1}\Bigg|_0^t\right]$$

$$e^{t}\cos(t)*e^{-t}=e^{-t}\left[\frac{e^{2t}\left(2\cos(t)+sen(t)\right)}{5}-\frac{2}{5}\right]$$

$$e^{t}\cos(t)*e^{-t}=\frac{e^{t}\left(2\cos(t)+sen(t)\right)}{5}-\frac{2}{5}e^{-t}$$

$$\mathscr{L}^{-1}\left\{\frac{s-1}{(s+1)\left[(s-1)^2+1\right]}\right\}=\frac{e^{t}\left(2\cos(t)+sen(t)\right)}{5}-\frac{2}{5}e^{-t}$$

Transformada de una Integral

$$\begin{aligned}\mathscr{L}\{f(t)*1\}&=\mathscr{L}\{f(t)\}\mathscr{L}\{1\}\\&=\mathscr{L}\{1\}\mathscr{L}\{f(t)\}\ \text{El producto es conmutativo.}\\&=\frac{1}{s}\mathscr{L}\{f(t)\}\end{aligned}$$

$$\mathscr{L}\{f(t)*1\}=\frac{1}{s}F(s)$$

$$\mathscr{L}\left\{\int_0^t f(\tau)d\tau\right\}=\frac{F(s)}{s}$$

$$\mathscr{L}\left\{\int_0^t f(\tau)d\tau\right\}=\frac{F(s)}{s}$$

La Forma Inversa

$$\mathscr{L}^{-1}\left\{\frac{F(s)}{s}\right\}=\int_0^t f(\tau)d\tau$$

Ejemplo: Hallar $\mathscr{L}\left\{ \int_0^t e^{\tau} d\tau \right\}$

Solución

$$\mathscr{L}\left\{ \int_0^t e^{\tau} d\tau \right\} = \frac{1}{s}\mathscr{L}\left\{e^t\right\}$$
$$= \frac{1}{s}\left(\frac{1}{s-1} \right)$$
$$= \frac{1}{s(s-1)}$$

Ejemplo: Hallar $\mathscr{L}\left\{\int_0^t e^{-\tau}\cos(\tau)d\tau\right\}$

Solución

$$\mathscr{L}\left\{\int_0^t e^{-\tau}\cos(\tau)d\tau\right\} = \frac{1}{s}\mathscr{L}\left\{e^{-t}\cos(t)\right\}$$
$$= \frac{1}{s}\mathscr{L}\left\{\cos(t)\right\}\Big|_{s\to s+1}$$
$$= \frac{1}{s}.\left(\frac{s}{s^2+1}\Big|_{s\to s+1}\right)$$
$$\mathscr{L}\left\{\int_0^t e^{-\tau}\cos(\tau)d\tau\right\} = \frac{1}{s}.\frac{s+1}{(s+1)^2+1}$$

Ejemplo: Hallar $\mathscr{L}\left\{ t\int_0^t sen(\tau)d\tau \right\}$

Solución

$$\mathscr{L}\left\{ t\int_0^t sen(\tau)d\tau \right\} = (-1)^1 \frac{d}{ds}\mathscr{L}\left\{ \int_0^t sen(\tau)d\tau \right\}$$
$$= -\frac{d}{ds}\left(\frac{1}{s}\mathscr{L}\left\{ sen(t)\right\}\right)$$
$$= -\frac{d}{ds}\left(\frac{1}{s}\bullet\frac{1}{s^2+1}\right)$$
$$= -\frac{d}{ds}\left(\frac{1}{s^3+s}\right)$$
$$= -\frac{(-1)(3s^2+1)}{\left(s^3+s\right)^2}$$
$$= \frac{3s^2+1}{\left(s^3+s\right)^2}$$
$$\mathscr{L}\left\{ t\int_0^t sen(\tau)d\tau \right\} = \frac{3s^2+1}{s^2(s^2+1)^2}$$

Teorema: (Transformada de Una Función Periódica)

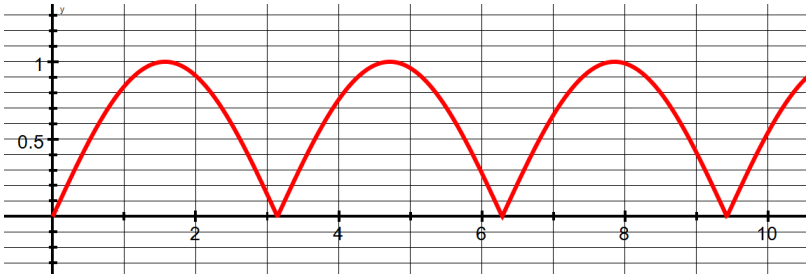
Si $f(t)$ es continua por tramos en $[0,+\infty[$, de orden exponencial y periódica con período T, entonces:

$$\mathscr{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)dt$$

Recordemos que si f es una función periódica de período T > 0 entonces
$f(t+T) = f(t)$

Ejemplo:

Encontrar la transformada de Laplace de
$f(t) = |sen(t)|$



Esta función es periódica de periodo $T = \pi$

Solución

$$\mathscr{L}\{f(t)\} = \frac{1}{1-e^{-\pi s}} \int_0^\pi e^{-st} f(t)dt$$

$$= \frac{1}{1-e^{-\pi s}} \left[\int_0^\pi e^{-st} sen(t)dt \right]$$

$$= \frac{1}{1-e^{-\pi s}} \left[\int_0^\pi e^{-st} sen(t)dt \right]$$

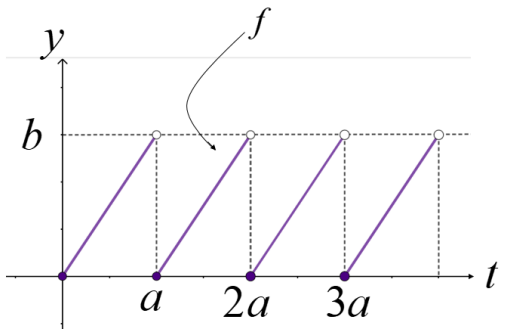
$$= \frac{1}{1-e^{-\pi s}} \left[e^{-st} \frac{-s\, sen(t) - \cos(t)}{(-s)^2+1} \right]_0^\pi$$ Utilizando integración por partes.

$$= \frac{1}{1-e^{-\pi s}} \left[\frac{-e^{-\pi s}(-1)}{s^2+1} - \frac{-1}{s^2+1} \right]$$

$$= \frac{e^{-\pi s}+1}{(1-e^{-\pi s})(s^2+1)}$$

Ejemplo

Encontrar la transformada de Laplace de la función f que se muestra en la figura



Solución

$T = a$

$$f(t) = \begin{cases} \dfrac{b}{a}t, & 0 \le t < a \\ \dfrac{b}{a}(t-a), & a \le t < 2a \\ \vdots \end{cases}$$

$$\mathscr{L}\{f(t)\} = \frac{1}{1-e^{-sT}}\int_0^T e^{-st} f(t)dt$$

$$\mathscr{L}\{f(t)\} = \frac{1}{1-e^{-as}}\int_0^a e^{-st}\left(\frac{b}{a}t\right)dt$$

$$\mathscr{L}\{f(t)\} = \frac{b}{a(1-e^{-as})}\int_0^a e^{-st}tdt$$

$u = t$

$du = dt$

$dv = e^{-st}dt$

$v = \int e^{-st}dt = -\frac{1}{s}e^{-st}$

$$\mathscr{L}\{f(t)\} = \frac{b}{a(1-e^{-as})}\left[-\frac{e^{-st}t}{s}\Big|_0^a - \int_0^a -\frac{e^{-st}}{s}dt\right]$$

$$\mathscr{L}\{f(t)\} = \frac{b}{a(1-e^{-as})}\left[-\frac{e^{-st}t}{s}\Big|_0^a - \frac{1}{s^2}e^{-st}\Big|_0^a\right]$$

$$\mathscr{L}\{f(t)\} = \frac{b}{a(1-e^{-as})}\left[-\frac{ae^{-as}}{s} + 0 - \frac{1}{s^2}\left(e^{-as}-1\right)\right]$$

$$\mathscr{L}\{f(t)\} = \frac{b}{a(1-e^{-as})}\left[-\frac{ae^{-as}}{s} + \frac{1}{s^2}\left(1-e^{-as}\right)\right]$$

$$\boxed{\mathscr{L}\{f(t)\} = \frac{b}{as^2} - \frac{be^{-as}}{s(1-e^{-as})}}$$

Transformada de Laplace de una función periódica utilizando la función escalón unitaria (en este caso no es necesario utilizar integración)

$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-sT}}\right]\mathscr{L}\{g(t)\}$, donde $g(t)$ es la función definida en un periodo y cero para cualquier otro valor.

Ejemplo

Encontrar la transformada de Laplace de

$f(t) = |sen(t)|$

Solución

Esta transformada ya ha sido resuelta, utilizando integrales ahora lo haremos utilizando la función escalón unitaria

$f(t) = |sen(t)|$

$T = \pi$

$$g(t) = \begin{cases} sen(t), & 0 \le t \le \pi \\ 0, & t \ge \pi \end{cases}$$

$$\mathscr{L}\{|sen(t)|\} = \left[\frac{1}{1-e^{-\pi s}}\right]\mathscr{L}\{g(t)\}, \qquad g(t) = sen(t)u(t) - sen(t)u(t-\pi)$$

$$\mathscr{L}\{|sen(t)|\} = \left[\frac{1}{1-e^{-\pi s}}\right]\mathscr{L}\{sen(t) - sen(t)u(t-\pi)\}$$

$$\mathscr{L}\{|sen(t)|\} = \left[\frac{1}{1-e^{-\pi s}}\right]\left[\frac{1}{s^2+1} - e^{-\pi s}\mathscr{L}\{sen(t+\pi)\}\right]$$

$$\mathscr{L}\{|sen(t)|\} = \left[\frac{1}{1-e^{-\pi s}}\right]\left[\frac{1}{s^2+1} - e^{-\pi s}\mathscr{L}\{sen(t)\cos(\pi) + sen(\pi)\cos(t)\}\right]$$

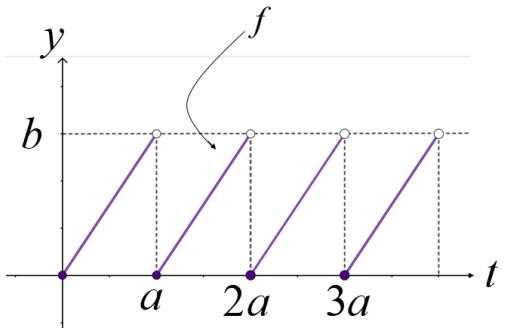
$$\mathscr{L}\{|sen(t)|\} = \left[\frac{1}{1-e^{-\pi s}}\right]\left[\frac{1}{s^2+1} - e^{-\pi s}\mathscr{L}\{-sen(t)\}\right]$$

$$\mathscr{L}\{|sen(t)|\} = \left[\frac{1}{1-e^{-\pi s}}\right]\left[\frac{1}{s^2+1} + e^{-\pi s}\frac{1}{s^2+1}\right]$$

$$\mathscr{L}\{|sen(t)|\} = \frac{e^{-\pi s}+1}{\left(1-e^{-\pi s}\right)\left(s^2+1\right)}$$

Ejemplo

Encontrar la transformada de Laplace de función f que se muestra en la figura



Solución

$T = a$

$$f(t) = \begin{cases} \frac{b}{a}t, 0 \le t < a \\ \frac{b}{a}(t-a), a \le t < 2a \\ \vdots \end{cases}$$

$$g(t) = \begin{cases} \frac{b}{a}t, & 0 \le t < a \\ 0, & t \ge a \end{cases}$$

$$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-as}}\right]\mathscr{L}\{g(t)\}, \qquad g(t) = \frac{b}{a}t\,u(t) - \frac{b}{a}t\,u(t-a)$$

$$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-as}}\right]\mathscr{L}\left\{\frac{b}{a}t - \frac{b}{a}t\,u(t-a)\right\}$$

$$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-as}}\right]\left[\frac{b}{as^2} - \frac{b}{a}e^{-as}\mathscr{L}\{t+a\}\right]$$

$$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-as}}\right]\left[\frac{b}{as^2} - \frac{b}{a}e^{-as}\left(\frac{1}{s^2} + \frac{a}{s}\right)\right]$$

$$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-as}}\right]\left[\frac{b}{as^2} - \frac{b}{as^2}e^{-as} - \frac{b}{s}e^{-as}\right]$$

$$\mathscr{L}\{f(t)\} = \left[\frac{1}{1-e^{-as}}\right]\left[\frac{b}{as^2}\left(1-e^{-as}\right) - \frac{b}{s}e^{-as}\right]$$

$$\boxed{\mathscr{L}\{f(t)\} = \frac{b}{as^2} - \frac{be^{-as}}{s\left(1-e^{-as}\right)}}$$

Aplicaciones

Se resolverán algunas ecuaciones diferenciales haciendo uso de los teoremas de traslación, convolución y otros.

Ejemplo

Resolver: $y''-4y'+4y=t^3e^{2t}, \quad y(0)=0, \quad y'(0)=0$

Solución

Aplicamos la transformada de Laplace en ambos miembros de la igualdad:

$$\mathscr{L}\{y''-4y'+4y\}=\mathscr{L}\{t^3e^{2t}\}$$

$$\mathscr{L}\{y''\}-4\mathscr{L}\{y'\}+4\mathscr{L}\{y\}=(-1)^3\frac{d^3}{ds^3}\mathscr{L}\{e^{2t}\}, \text{ propiedad de linealidad}$$

$$\left[s^2Y(s)-sy(0)-y'(0)\right]-4\left[sY(s)-y(0)\right]+4Y(s)=-\frac{d^3}{ds^3}\left[\frac{1}{s-2}\right]$$

$$s^2Y(s)-4sY(s)+4Y(s)=-\frac{d^3}{ds^3}\left[(s-2)^{-1}\right]$$

$$\left(s^2-4s+4\right)Y(s)=-\frac{d^2}{ds^2}\left[-(s-2)^{-2}\right]$$

$$(s-2)^2\,Y(s)=-\frac{d}{ds}\left[2(s-2)^{-3}\right]$$

$$(s-2)^2\,Y(s)=-\left[-6(s-2)^{-4}\right]$$

$$(s-2)^2\,Y(s)=\frac{6}{(s-2)^4}$$

$$Y(s)=\frac{6}{(s-2)^6}$$

$$\mathscr{L}^{-1}\{Y(s)\}=\mathscr{L}^{-1}\left\{\frac{6}{(s-2)^6}\right\}$$

$$y(t)=6\mathscr{L}^{-1}\left\{\frac{1}{(s)^6}\Big|_{s\to s-2}\right\}$$

$$y(t)=6e^{2t}\mathscr{L}^{-1}\left\{\frac{1}{(s)^6}\right\}$$

$$y(t)=\frac{6}{5!}e^{2t}\mathscr{L}^{-1}\left\{\frac{5!}{(s)^6}\right\}$$

$\boxed{y(t)=\frac{1}{20}e^{2t}t^5}$ Solucion del PVI

Ejemplo

Resolver: $y' + y = f(t), \quad y(0) = 0$

Donde $f(t) = \begin{cases} 1, & 0 \le t < 1 \\ -1, & t \ge 1 \end{cases}$

Solución

Transformamos la función seccionada en términos de funciones escalones unitarias.

$$f(t) = 1\left[U(t) - U(t-1)\right] - 1U(t-1)$$
$$f(t) = 1 - 2U(t-1)$$

$$y' + y = f(t)$$
$$y' + y = 1 - 2U(t-1)$$

Aplicamos la transformada de Laplace en ambos miembros:

$$\mathscr{L}\{y' + y\} = \mathscr{L}\{1 - 2U(t-1)\}$$
$$\mathscr{L}\{y'\} + \mathscr{L}\{y\} = \mathscr{L}\{1\} - 2\mathscr{L}\{U(t-1)\}$$
$$sY(s) - y(0) + Y(s) = \frac{1}{s} - 2\left(\frac{e^{-s}}{s}\right)$$
$$sY(s) + Y(s) = \frac{1}{s} - \frac{2e^{-s}}{s}$$
$$(s+1)Y(s) = \frac{1}{s} - \frac{2e^{-s}}{s}$$
$$Y(s) = \frac{1}{s(s+1)} - \frac{2e^{-s}}{s(s+1)}$$

$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} = \frac{1}{s} - \frac{1}{s+1}$$
$$1 = A(s+1) + Bs$$
$$s = 0 \rightarrow A = 1$$
$$s = -1 \rightarrow B = -1$$

$$Y(s)=\frac{1}{s}-\frac{1}{s+1}-2e^{-s}\left(\frac{1}{s}-\frac{1}{s+1}\right)$$

$$Y(s)=\frac{1}{s}-\frac{1}{s+1}-\frac{2e^{-s}}{s}+\frac{2e^{-s}}{s+1}$$

$$\mathscr{L}^{-1}\{Y(s)\}=\mathscr{L}^{-1}\left\{\frac{1}{s}\right\}-\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}-\mathscr{L}^{-1}\left\{\frac{2e^{-s}}{s}\right\}+\mathscr{L}^{-1}\left\{\frac{2e^{-s}}{s+1}\right\}$$

$$y(t)=1-e^{-t}-2U(t-1)+2\mathscr{L}^{-1}\left\{\frac{e^{-s}}{s+1}\right\}$$

$$y(t)=1-e^{-t}-2U(t-1)+2e^{-(t-1)}U(t-1)$$

$$y(t)=1-e^{-t}+\left(-2+2e^{-(t-1)}\right)U(t-1)$$

$$\mathscr{L}^{-1}\left\{\frac{e^{-s}}{s+1}\right\}=e^{-(t-1)}U(t-1)$$

$$\downarrow \qquad\qquad \uparrow$$

$$\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}=e^{-t}$$

Faltaría expresar esta función como función seccionada

$$y(t)=1-e^{-t}+\left(-2+2e^{-(t-1)}\right)U(t-1)$$

	0	1	$+\infty$
$1-e^{-t}$	$1-e^{-t}$	$1-e^{-t}$	
$\left(-2+2e^{-(t-1)}\right)U(t-1)$	0	$-2+2e^{-(t-1)}$	
$y(t)$	$1-e^{-t}$	$-1+2e^{-(t-1)}-e^{-t}$	

$$y(t)=\begin{cases}1-e^{-t}, & 0\le t<1\\ -1+\left(2e-1\right)e^{-t}, & t\ge 1\end{cases}$$

Ejemplo

Determine $f(t)$, Si $f(t)=\cos(t)+\int_0^t e^{-\tau} f(t-\tau)d\tau$

Solución

Aplicando la transformada de Laplace en ambos miembros de la ecuación.

$$\mathscr{L}\{f(t)\}=\mathscr{L}\{\cos(t)\}+\mathscr{L}\left\{\int_0^t e^{-\tau} f(t-\tau)d\tau\right\}$$

$$F(s)=\frac{s}{s^2+1}+\mathscr{L}\left\{e^{-t}*f(t)\right\}$$

$$F(s)=\frac{s}{s^2+1}+\mathscr{L}\left\{e^{-t}\right\}\mathscr{L}\{f(t)\}$$

$$F(s)=\frac{s}{s^2+1}+\frac{1}{s+1}F(s)$$

$$F(s)-\frac{1}{s+1}F(s)=\frac{s}{s^2+1}$$

$$\left(1-\frac{1}{s+1}\right)F(s)=\frac{s}{s^2+1}$$

$$\left(\frac{s+1-1}{s+1}\right)F(s)=\frac{s}{s^2+1}$$

$$\left(\frac{s}{s+1}\right)F(s)=\frac{s}{s^2+1}$$

$$F(s)=\frac{s(s+1)}{s(s^2+1)}$$

$$F(s)=\frac{s+1}{s^2+1}$$

$$\mathscr{L}^{-1}\{F(s)\}=\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\right\}+\mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$\boxed{f(t)=\cos(t)+sen(t)}$$